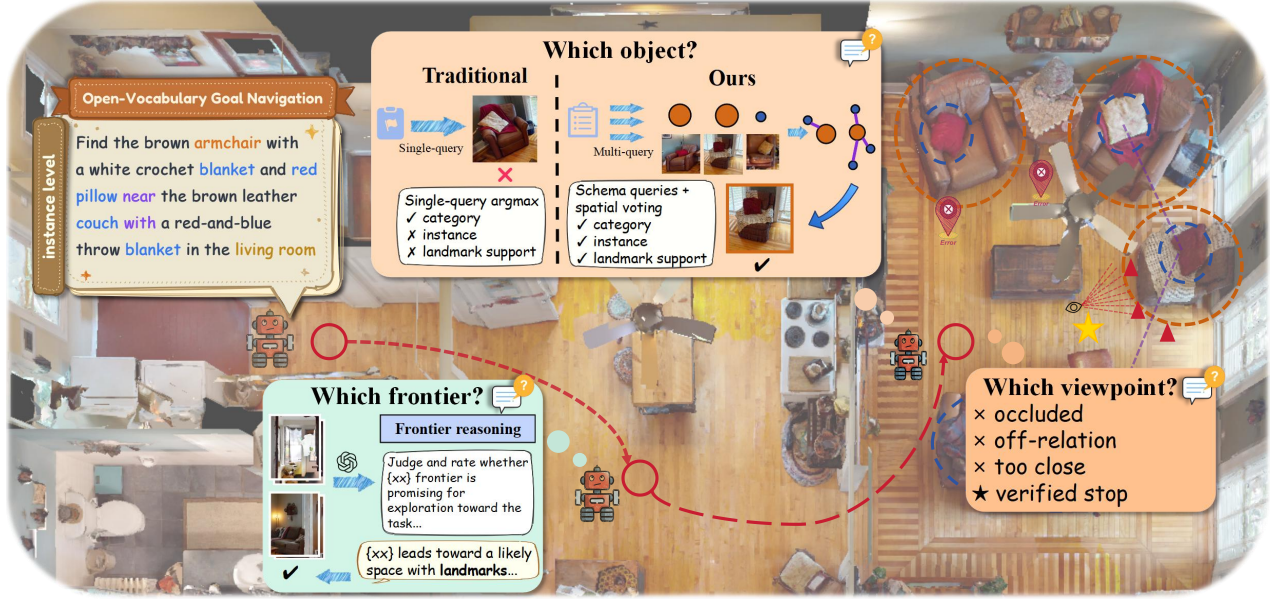


LANDER-Nav: Progressive Landmark-Schema Grounding for Multi-Granularity Open-Vocabulary Navigation

Anonymous submission



Abstract

Language-conditioned goal navigation is extending beyond category-level search toward multi-granularity goals disambiguated by landmarks, context, and spatial relations. However, current methods encode instructions jointly, conflating these constraints and often causing agents to bind the wrong same-category instance or stop at an unverifiable viewpoint. We disentangle each instruction into a landmark schema comprising target, anchor, context, and relation fields, and propose LANDER-Nav, a progressive grounding framework that preserves these role-specific constraints throughout navigation. Under this schema, LANDER-Nav decomposes navigation into three coupled decisions: Evidence Grounding decides where to explore by selecting frontiers likely to reveal missing evidence; Entity Grounding decides which entity to bind through landmark-anchored spatial voting over object memory; and Endpoint Grounding decides where to stop by choosing an executable viewpoint that verifies the required relation. Extensive experiments on LangMap and HM3D-OVON demonstrate that LANDER-Nav consistently improves language-conditioned navigation, particularly on context- and relation-dependent goals, while improving category-level open-vocabulary performance.

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Introduction

Embodied open-vocabulary navigation requires robotic agents to locate language-specified goals in unseen environments. Early benchmarks (Ramakrishnan et al. 2021; Chang et al. 2017; Anderson et al. 2018; Qi et al. 2020; Ku et al. 2020; Zhu et al. 2021) primarily focus on route following, predefined object categories, or single referring targets. Recent open-vocabulary goal navigation methods (Gadre et al. 2023; Zhou et al. 2023; Yokoyama et al. 2024; Padilla-Cerdio et al. 2026) relax the target space to free-form language goals. In practice, however, targets are rarely specified by category names alone, but are instead identified through visual attributes, surrounding landmarks, scene context, and spatial relations. To capture this complexity, LangMap (Miao et al. 2026a) formalizes the problem as multi-granularity open-vocabulary navigation across Object, Room, Region, and Instance goals, where agents localize target objects under increasingly specific contextual constraints.

Existing open-vocabulary goal navigation methods (Gadre et al. 2023; Zhou et al. 2023; Yokoyama et al. 2024; Padilla-Cerdio et al. 2026) typically construct semantic maps or object-centric memories, and use vision-language models for frontier selection, candidate ranking, and semantic reasoning. Although these approaches improve open-vocabulary retrieval and exploration, their decisions often rely on a single target embedding or a holistic sentence-level similarity

score. This design makes it challenging to explicitly disentangle and continuously reason about the target, reference landmarks, contextual cues, and spatial relations. For instance, given a query “the lamp on the table to the left of the bed,” sentence-level matching may retrieve a lamp without identifying the specific instance jointly constrained by the table and the bed. Multi-granularity open-vocabulary navigation therefore calls for structured language constraints that persist across exploration, entity binding, and stop verification.

Motivated by the role of landmarks in human localization and spatial organization (Epstein and Vass 2014; Epstein et al. 2017), we reinterpret landmarks as executable language roles rather than auxiliary descriptors. From this perspective, a context-dependent navigation goal then requires the agent to progressively resolve three coupled questions. First, before the target or its supporting evidence has been observed, *which frontier* is most likely to reveal the missing target, anchor, or contextual cues? Second, when multiple category-consistent objects coexist in memory, *which entity* satisfies the target description jointly with the required landmarks and spatial relations? Third, once an entity is bound, *which reachable viewpoint* enables the agent to observe the target under the intended target–anchor configuration? Together, these questions suggest that exploration, object binding, and stopping should be governed by a unified structured language representation.

Based on this insight, we propose LANDER-Nav, a progressive landmark-schema grounding framework for multi-granularity open-vocabulary navigation. LANDER-Nav parses each natural-language instruction into a **landmark schema** that explicitly represents the target, reference landmarks, contextual regions, and spatial relations, and maintains this structure across three grounding stages. Within our framework, *Evidence Grounding* selects frontiers likely to reveal missing schema evidence; *Entity Grounding* binds the target entity under joint target–anchor–context–relation constraints; *Endpoint Grounding* employs Landmark-Ray Endpoint Search to identify a reachable, visible, and relation-consistent stopping viewpoint based on target–anchor geometry. Extensive experiments on LangMap and HM3D-OVON demonstrate that LANDER-Nav consistently outperforms strong baselines, with pronounced gains on context- and relation-dependent goals while also improving category-level open-vocabulary performance. These results suggest that explicit landmark-schema grounding effectively supports multi-granularity navigation while preserving strong performance on category-level open-vocabulary search.

Our contributions are summarized as follows:

- We introduce LANDER-Nav, which formulates multi-granularity open-vocabulary navigation as a progressive landmark-schema grounding framework, and use a unified landmark schema to coordinate frontier selection, target binding, and stopping-viewpoint search.
- We propose a landmark-anchored object binding reasoning framework. We introduce a spatial-voting-based reasoning framework that uses reference landmarks and spatial relations in the task to vote for and evaluate target candidates, addressing the limitation that conventional

methods compress the task into a single query representation and therefore lose role-level semantic constraints.

- We demonstrate that explicit landmark-schema grounding yields consistent performance gains and strong generalization in open-vocabulary navigation. Extensive experiments on LangMap and HM3D-OVON show improved SR/SPL for multi-granularity goals, while maintaining robust performance on category-level open-vocabulary navigation.

Related Work

Goal-Oriented Navigation

Goal-oriented navigation requires an agent to actively search for and reach a target location according to a language or multimodal goal. Early vision-language navigation mainly focuses on route following, aligning natural-language instructions with indoor viewpoint sequences or continuous action trajectories (Anderson et al. 2018; Ku et al. 2020). REVERIE and SOON then move the problem toward remote object localization and scenario-description-driven goal navigation, where the agent must actively identify the language-referred object in a partially observable environment (Qi et al. 2020; Zhu et al. 2021). These tasks advance language-conditioned navigation, but their goals are still usually organized around routes, object categories, or single referring expressions.

Open-vocabulary goal navigation further relaxes the target set, enabling agents to find objects from categories that are not predefined (Gadre et al. 2023; Zhou et al. 2023). VLFM and OpenFrontier use vision-language models to select target-relevant frontiers or exploration directions, connecting open-vocabulary goals to online navigation decisions (Yokoyama et al. 2024; Padilla-Cerdio et al. 2026). More recent work such as GOAT-Bench, PSL, and Uni-NaVid attempts to unify different goal modalities or navigation tasks and reduce task-specific designs (Khanna et al. 2024; Sun et al. 2024; Zhang et al. 2025). LangMap further organizes open-vocabulary goals into Object, Room, Region, and Instance levels, better matching how real users describe navigation targets from coarse to specific granularity (Miao et al. 2026a).

Existing methods usually enhance target search with semantic maps, object memories, or trajectory histories (Chen et al. 2024; Ao et al. 2026). Some methods use large models for high-level planning, candidate judgment, or uncertainty handling (Xue et al. 2025; Shinnick and Miao 2026). Long-horizon memory and structured history representations further improve navigation stability in complex environments (Ren et al. 2025; Zeng et al. 2026; Zhou et al. 2025; Yin et al. 2025). However, these methods often compress the language goal into a full-sentence query, a target embedding, or a single candidate score, making it difficult to continuously preserve the role differences among the target, reference landmarks, contextual regions, and spatial relations. For multi-granularity goals, this compression weakens the agent’s understanding of instance-level semantic constraints.

Open-Vocabulary Navigation Decision

Open-vocabulary navigation decisions are usually built on matching a language query to candidates. CoW uses open-vocabulary models to localize target objects and combines them with an exploration strategy to decide when to move toward the target (Gadre et al. 2023). ESC estimates where the target is more likely to appear through commonsense constraints (Zhou et al. 2023). VLFM and OpenFrontier align target text with visual observations or frontier representations, and select the next navigation direction with relevance scores (Yokoyama et al. 2024; Padilla-Cerdio et al. 2026). These methods turn open-vocabulary goals into executable decisions, but the final judgment still mainly depends on the overall match between target semantics and candidates.

Later work introduces richer spatial representations to support open-vocabulary decisions. VLMaps and Concept-Graphs write language semantics into maps or 3D scene graphs for subsequent query and planning (Huang et al. 2023; Gu et al. 2024). 3D-Mem and MSGNav further store multi-view memory or multimodal scene graphs to strengthen visual and spatial evidence in candidate retrieval (Yang et al. 2025; Huang et al. 2026). Recent approaches further explore physically executable representations, 3D grounding, and active exploration for open-vocabulary navigation (Miao et al. 2026b; Li et al. 2025; Zhu et al. 2025). Although richer scene representations provide useful environmental evidence, navigation decisions may still collapse complex instructions into a single query-candidate matching score.

We propose **landmark-anchored object binding** to mitigate this limitation. We use key landmarks and spatial relations as explicit evidence in candidate evaluation, and judge whether a target candidate satisfies both target semantics and landmark constraints through spatial voting. In this way, the final navigation decision no longer depends only on single-query similarity, but is jointly determined by the target, reference landmarks, and spatial relations, reducing the loss of role-level semantic constraints during candidate binding.

Method

Framework Overview

We study multi-granularity open-vocabulary navigation. Given a natural-language goal x , an agent must continuously perceive an unknown environment, update its memory, and decide whether to keep exploring or navigate to the target according to its current scene knowledge. The overall pipeline of LANDER-Nav is shown in Fig. 1. It contains three core parts: first parsing the landmark schema in language, then building online scene memory, and finally grounding frontier, object, and viewpoint decisions based on this memory.

Given a natural-language instruction x , LANDER-Nav first uses a vision-language model to parse its landmark schema:

$$\mathcal{S}_x = (g, \mathcal{A}, \mathcal{C}, \mathcal{R}). \quad (1)$$

Here, g denotes the target-object description; $\mathcal{A} = \{a_m\}_{m=1}^{MA}$ denotes the set of reference landmarks; $\mathcal{C}$ denotes room, region, or layout context; and $\mathcal{R}$ denotes spatial relations between the target and reference landmarks, such as near, with,

on, and in-room. For category-level or short instructions, $\mathcal{A}$, $\mathcal{C}$, or $\mathcal{R}$ may be empty. This schema decomposes the original language goal into four types of verifiable evidence, target, anchor, context, and relation, providing a unified condition for subsequent exploration, instance binding, and stopping-viewpoint selection.

During navigation, the agent maintains scene memory online. The memory state is denoted as:

$$\mathcal{Z}_t = (\mathcal{O}_t, \mathcal{M}_t, \mathcal{F}_t), \quad (2)$$

where $\mathcal{O}_t$ is object-level memory, $\mathcal{M}_t$ is the top-down occupancy and exploration map, and $\mathcal{F}_t$ is the current frontier set. For each RGB-D observation and agent pose $z_t = (I_t, D_t, P_t)$, the system first partitions the image into local regions, and uses depth and pose to align these regions into the global 3D coordinate frame. The local regions are then encoded as object candidates and matched and fused with historical object entries, yielding continuously updated object memory:

$$\mathcal{O}_t = \{o_i^t\}_{i=1}^{N_t}, \quad o_i^t = (b_i^t, m_i^t, f_i^t, v_i^t, s_i^t). \quad (3)$$

Here, b_i^t denotes the 3D bounding box of the object, m_i^t denotes the segment-level mask from SAM, f_i^t denotes visual-geometric features, v_i^t denotes open-vocabulary semantic embeddings, and s_i^t denotes detection or fusion confidence. Local observations from different viewpoints are accumulated into object-centered long-term memory, allowing the system to perform open-vocabulary retrieval and spatial-relation reasoning without relying on complete scene annotations.

Meanwhile, the system continuously updates the top-down map $\mathcal{M}_t$ from depth observations and poses, dividing space into explored, traversable, obstacle, and unknown regions. The frontier set $\mathcal{F}_t = \{f_j^t\}_{j=1}^{M_t}$ is generated from the boundary between explored and unknown regions, representing potential next exploration targets.

Based on the schema $\mathcal{S}_x$ and online memory $\mathcal{Z}_t$, LANDER-Nav maintains a schema-support state $\Gamma_t(\mathcal{S}_x)$ at each high-level decision step, indicating which evidence among target, anchor, context, and relation is already supported in the current memory and which evidence is still missing. The system then branches between exploration and target navigation according to object-candidate confidence, schema-support state, and navigability conditions. If the current memory is insufficient to bind the target reliably, the system explores. If the target and its supporting landmarks have formed stable evidence, the system further selects the target object and stopping viewpoint.

Specifically, LANDER-Nav contains three grounding modules aligned with Fig. 1. First, **Evidence Grounding** answers *Which frontier?* by selecting the exploration direction in the current frontier set that is most likely to reveal missing schema evidence. Second, **Entity Grounding** answers *Which object?* by binding a target instance in object memory using target, anchor, context, and relation evidence. Third, **Endpoint Grounding** answers *Which viewpoint?* by converting the bound target into a reachable, visible, and relation-verifiable final stopping viewpoint.

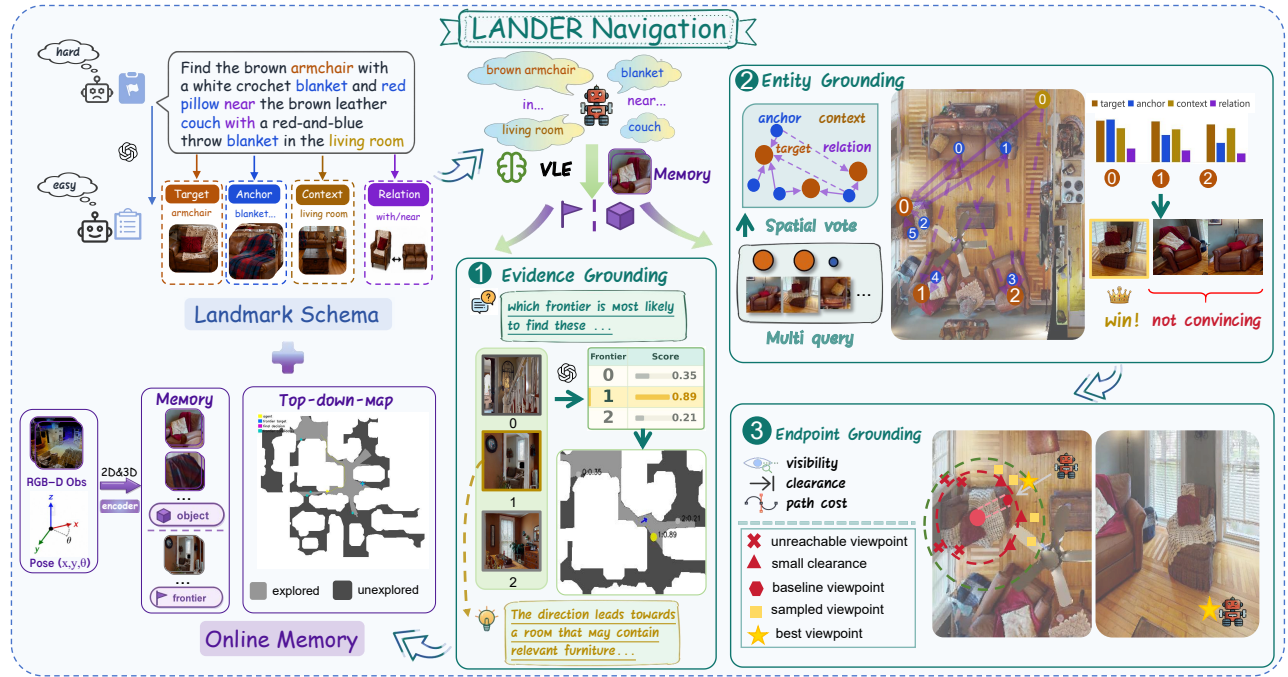


Figure 1: Framework of LANDER-Nav. The instruction is parsed into a landmark schema with target, anchor, context, and relation fields. Evidence Grounding selects frontiers that may expose missing schema evidence. Entity Grounding binds the target entity under target-anchor-context-relation constraints. Endpoint Grounding uses Landmark-Ray Endpoint Search to produce a reachable, visible, and relation-verifiable stopping viewpoint.

The three decisions share a unified candidate-selection form:

$$\hat{y}_m = \arg \max_{y \in \mathcal{Y}_m(t)} S_m(y | \mathcal{S}_x, \mathcal{Z}_t), \quad m \in \{F, O, V\}. \quad (4)$$

Here, F , O , and V correspond to frontier, object, and viewpoint. $\mathcal{Y}_F(t) = \mathcal{F}_t$ denotes frontier candidates, $\mathcal{Y}_O(t) = \mathcal{O}_t$ denotes object-memory candidates, and $\mathcal{Y}_V(t)$ denotes navigable viewpoint candidates generated around the bound target. The scoring function S_m is conditioned on the current schema $\mathcal{S}_x$ and online memory $\mathcal{Z}_t$, but uses different evidence for different candidate types. Frontier scoring emphasizes whether the candidate leads to a relevant region; object scoring emphasizes semantic and spatial-relation consistency; viewpoint scoring emphasizes reachability, visibility, clearance, and verifiability of the target-landmark relation.

The whole pipeline uses only current and past RGB-D observations, agent poses, and the online memory constructed from them. LANDER-Nav can progressively collect landmark evidence during exploration, and complete schema-level target binding and reliable navigation stopping once the evidence becomes sufficient.

Evidence Grounding: Which Frontier?

Evidence Grounding determines whether the next exploration direction may reveal evidence missing from the current landmark schema. Each frontier f_j is converted into a currently observable task-facing direction. According to the unsatisfied target, anchor, or context fields in $\Gamma_t(\mathcal{S}_x)$, the

system evaluates whether this direction leads to a relevant region.

For a candidate frontier f_j , let the agent's horizontal position be $p_{t,xz}$, and the frontier horizontal position be $f_{j,xz}$. Its direction vector is:

$$d_j = \frac{f_{j,xz} - p_{t,xz}}{\|f_{j,xz} - p_{t,xz}\|_2}. \quad (5)$$

If the current panorama contains K yaw-aligned views $\{I_t^k\}_{k=1}^K$, and each view corresponds to direction ψ_k , the system selects the frontier-facing view whose orientation is closest to d_j :

$$k_j = \arg \min_k |\text{wrap}(\angle(d_j) - \psi_k)|, \quad I_j = I_t^{k_j}. \quad (6)$$

I_j is the visual evidence observed when the agent faces f_j from its current position. The vision-language scorer receives I_j , instruction x , schema $\mathcal{S}_x$, frontier metadata, and a current memory summary, and returns:

$$r_j = (c_j^g, c_j^A, c_j^C, c_j^-, \kappa_j), \quad (7)$$

where c_j^g , c_j^A , and c_j^C denote target-context, anchor-context, and room/layout-context support, c_j^- denotes negative evidence, and κ_j is confidence. The task-facing score is:

$$H_F(f_j) = \beta_g c_j^g + \beta_A c_j^A + \beta_C c_j^C - \beta_- c_j^-. \quad (8)$$

For example, for the brown armchair instruction in the framework figure, a frontier leading to the living room and observing cues such as the couch, blanket, or pillow receives

a higher H_F . VLM scoring is executed at a fixed interval Δ_{vlm} , and covers the current candidate, the top- K_F frontiers in the ranking, and competing candidates whose score gap is smaller than τ_F . Δ_{vlm} , K_F , and τ_F are fixed in the experimental setup. Other candidates reuse the most recent valid score; if no historical score exists, the semantic term is set to zero. The final frontier score is:

$$S_F(f_j) = \alpha_F \text{Norm}(\ell_j^F) + \alpha_H \tilde{H}_F(f_j), \quad (9)$$

where ℓ_j^F denotes a geometry-aware exploration prior, and $\tilde{H}_F(f_j)$ is the fresh or cached task-facing score. The candidate output is:

$$\hat{f} = f_{\hat{j}}, \quad \hat{j} = \arg \max_{1 \leq j \leq M_t} S_F(f_j). \quad (10)$$

LANDER-Nav selects the schema-conditioned frontier only when the decision confidence is sufficient:

$$\text{Accept}_F(\hat{j}) = \mathbb{I} \left[S_F(f_{\hat{j}}) - S_F(f_b) \geq \eta_F \right] \mathbb{I} \left[\kappa_{\hat{j}} \geq \eta_{\kappa} \right], \quad (11)$$

where f_b denotes the geometry-prior frontier selected without schema-conditioned evidence. If the acceptance condition is not satisfied, the policy follows f_b .

Entity Grounding: Which Object?

Entity Grounding binds the target-role instance referred to by language in object memory, while reference landmarks, context, and relations participate as instance-disambiguation evidence through voting. For “the lamp on the nightstand beside the white bed,” the lamp is the target, and the nightstand and bed are landmarks supporting target selection. LANDER-Nav uses **role-conditioned spatial vote**: target, anchor, and context first produce evidence separately, relation constraints then check whether this evidence forms a consistent local configuration, and the target-role object is finally selected inside that configuration.

LANDER-Nav converts the schema into a role-query set $\mathcal{Q}_O$, including the full-sentence query q_{full} , the target query q_g , reference-landmark queries $\{q_a\}$, and context queries $\{q_c\}$. $\mathcal{R}$ is not used as an independent retrieval query; instead, it participates in region scoring and target-role reselection through footprint distance, role coverage, and local relation support. For each role query q and object o_i , the open-vocabulary scoring interface gives $\ell_{q,i} = s_{\theta}(q, o_i)$, and $p_{q,i}$ is obtained through softmax. The system keeps the top- K_O hits for each role and takes their union as the candidate evidence pool $\mathcal{C}_O$. This stage preserves possible targets and anchors, allowing landmarks to participate in instance disambiguation.

The system then builds a footprint-aware spatial graph on $\mathcal{C}_O$. The horizontal footprint distance between two objects is:

$$d_{\text{fp}}(i, k) = \max(0, \|u_i - u_k\|_2 - \rho_i - \rho_k). \quad (12)$$

If $d_{\text{fp}}(i, k) \leq \epsilon_O$, objects o_i and o_k are connected. The connected components $\{\mathcal{L}_m\}_{m=1}^L$ denote candidate landmark-consensus regions. For role query q , region evidence is aggregated by noisy-or:

$$E_q(\mathcal{L}_m) = 1 - \prod_{o_i \in \mathcal{L}_m} (1 - p_{q,i}). \quad (13)$$

The region score integrates role evidence, schema coverage, spatial compactness, and relation consistency, where $\mathcal{Q}_{\text{req}} \subseteq \mathcal{Q}_O$ denotes non-empty target, anchor, and context roles required by the current schema:

$$\begin{aligned} S_{\text{reg}}(\mathcal{L}_m) = & \sum_{q \in \mathcal{Q}_{\text{req}}} w_q \log(\epsilon + E_q(\mathcal{L}_m)) \\ & + \lambda_{\text{cov}} \text{Cov}_O(\mathcal{L}_m) + \lambda_{\text{cmp}} \text{Cmp}_O(\mathcal{L}_m) \\ & + \lambda_{\text{rel}} \text{Rel}_O(\mathcal{L}_m). \end{aligned} \quad (14)$$

Cov_O denotes the proportion of valid roles covered by the region, Cmp_O rewards compact local configurations, and Rel_O denotes local spatial support between the target and supporting-role objects. The current implementation mainly approximates this support with footprint distance and role co-occurrence. The highest-scoring connected component is selected as the best spatial-consensus region $\mathcal{L}^*$.

This region-level voting first merges target, anchor, and context evidence inside a local configuration, and then performs target-role selection. In the armchair example in the framework figure, the correct armchair, blanket, pillow, couch, and living-room context fall into the same region, so the region receives votes from multiple evidence sources. An armchair of the same category without these supporting objects receives a lower score.

Within $\mathcal{L}^*$, the system selects the target-role object:

$$\begin{aligned} \hat{o} = \arg \max_{o_i \in \mathcal{L}^*} & \left[\alpha_g \log(\epsilon + p_{g,i}) + \alpha_x \log(\epsilon + p_{\text{full},i}) \right. \\ & \left. + \alpha_R \text{Rel}_O(o_i, \mathcal{L}^*, \mathcal{R}) + \alpha_s \text{Norm}(\ell_i^O) \right]. \end{aligned} \quad (15)$$

Here, $p_{g,i}$ is the target-role probability, $p_{\text{full},i}$ is full-sentence support, and ℓ_i^O is the object-memory retrieval prior. The system accepts $\hat{o}$ only when region coverage, target confidence, region-score margin, and retrieval consistency reach their thresholds; otherwise, it falls back to the top retrieval candidate and records the rejection reason. In this way, anchor and context form spatial evidence, while the final navigation target still falls on the target role g .

Endpoint Grounding: Which Viewpoint?

Endpoint Grounding converts the bound target into an executable stopping viewpoint. The object center may lie inside the object, behind an occluder, or in an unreachable area; therefore, directly projecting the object center produces unstable stopping points. LANDER-Nav uses **Landmark-Ray Endpoint Search (LRES)**, generating direction-constrained candidate rays along the target-anchor relation and selecting a viewpoint that is reachable, visible, distance-appropriate, and relation-verifiable.

Let $T_{\hat{o}}$ be the observed point cloud of the target object $\hat{o}$, B_t be the non-target blocker point cloud, and c_g be the target center. If an effective anchor a exists and its center is c_a , the target-anchor semantic axis is:

$$\mathbf{u}_{ga} = \frac{c_g - c_a}{\|c_g - c_a\|}. \quad (16)$$

LRES generates a landmark-ray field along this semantic axis

and its small angular perturbations:

$$\mathcal{R}_{ga} = \{c_g + \rho \text{Rot}(\phi)\mathbf{u}_{ga} \mid \rho \in [r_{\min}, r_{\max}], \phi \in [-\alpha, \alpha]\}. \quad (17)$$

ρ controls the horizontal distance between the candidate viewpoint and the target, while ϕ controls the angular deviation from the target-anchor axis. The distance band only constrains the stopping distance to the target; the candidate direction is determined by the target-anchor axis. In implementation, LRES discretizes ρ and ϕ , and checks navigability, whether the point belongs to the same navigable connected component, and duplicate candidates for each raw ray sample. The candidate set is the intersection between ray fields from valid anchors and navigable free space:

$$\mathcal{V}_{\text{ray}} = \mathcal{F}_{\text{reachable}} \cap \bigcup_{a \in \mathcal{A}_{\text{valid}}} \mathcal{R}_{ga}. \quad (18)$$

$\mathcal{F}_{\text{reachable}}$ is given by the navmesh or path finder, and requires the candidate to be in the same navigable connected component as the current agent. For raw ray samples that are not on the navmesh, LRES snaps them to the nearest point in the same navigable component, and accepts only candidates whose snap distance is no larger than δ_{snap} . If the schema has no valid anchor, LRES uses the horizontal axis from the target center toward the current agent position as a degenerate ray, so category-level goals can still enter the same interface.

For a candidate viewpoint v , the system computes target visibility, anchor co-visibility, relation verifiability, clearance, and path cost. Target visibility is obtained by ray-occlusion checking from camera height $v + h_{\text{cam}}$ to sampled target points:

$$V_g(v) = \frac{1}{|T_s|} \sum_{t_n \in T_s} \mathbb{I}[\text{Ray}(v + h_{\text{cam}}, t_n) \cap B_t = \emptyset]. \quad (19)$$

The candidate viewpoint score is:

$$S_V(v) = \lambda_g V_g(v) + \lambda_a V_A(v) + \lambda_r Q_R(v) + \lambda_c C(v) - \lambda_p d_{\text{geo}}(p_t, v). \quad (20)$$

$V_A(v)$ denotes anchor visibility or co-visibility. The current implementation approximates $Q_R(v)$ as the joint score of target-anchor axis alignment and $V_A(v)$. $C(v)$ is the minimum horizontal distance to blockers, and d_{geo} is the geodesic cost from the current pose to the candidate viewpoint. The final viewpoint is:

$$\hat{v} = \arg \max_{v \in \mathcal{V}_{\text{ray}}} S_V(v). \quad (21)$$

Before selection, LRES requires target visibility, clearance, and anchor visibility to satisfy thresholds:

$$\text{Feasible}_V(v) = \mathbb{I}[V_g(v) \geq \gamma_g] \mathbb{I}[V_A(v) \geq \gamma_a] \mathbb{I}[C(v) \geq \gamma_c]. \quad (22)$$

If the instruction degenerates into a category-level goal, V_A and Q_R are treated as empty constraints. If no feasible ray candidate exists, the system falls back to a target-centered navigation point and re-decides after subsequent observation updates.

Experiments

Experimental Setup

We evaluate LANDER-Nav on LangMap and HM3D-OVON. LangMap covers four multi-granularity goal levels, Object, Room, Region, and Instance. HM3D-OVON is used to examine generalization on category-level open-vocabulary goal navigation. Both experiments follow the corresponding benchmark episode setup, success criterion, and validation split.

We use Success Rate (SR) and Success weighted by Path Length (SPL) as the main metrics:

$$\text{SR} = \frac{1}{N} \sum_{i=1}^N S_i. \quad (23)$$

$$\text{SPL} = \frac{1}{N} \sum_{i=1}^N S_i \frac{L_i^*}{\max(L_i, L_i^*)}. \quad (24)$$

Here, N is the number of evaluation tasks, $S_i \in \{0, 1\}$ indicates whether the i -th task is successfully completed, and L_i and L_i^* denote the agent's executed path length and the corresponding shortest path length. SR measures target-localization success, while SPL combines success with normalized path length and reflects path efficiency among successful episodes.

In multi-goal tasks, different subtasks have sequential dependency; failure on an earlier goal prevents later goals from being counted as sequentially successful. To measure the agent's continuous completion ability, we report Sequential Success Rate (SeqSR):

$$\text{SeqSR}@n = \frac{1}{N_{\text{ep}}} \sum_{i=1}^{N_{\text{ep}}} \mathbb{I}\left(\sum_{j=1}^{K_i} S_{i,j} \geq n\right). \quad (25)$$

Here, N_{ep} is the number of episodes, K_i is the number of subtasks in the i -th episode, and $S_{i,j} \in \{0, 1\}$ indicates whether the j -th subtask in the i -th episode is completed in order. In LangMap, each multi-goal episode contains $K_i = 5$ subtasks. We denote the proportion of episodes completing all five subtasks as SeqSR, namely SeqSR@5, and the proportion completing at least four subtasks as SeqSR-4, namely SeqSR@4.

In implementation, we use a 0.25 m geodesic distance as the success threshold, and each LangMap subtask is allowed at most 500 steps. LANDER-Nav uses GPT-4o-mini as the default VLM for semantic scoring, and keeps component weights and thresholds fixed across all experiments.

Main Results on LangMap

'Table 1 shows that LANDER-Nav achieves the highest Overall SR/SPL under the detailed setting, reaching **45.5 / 28.6**. The performance advantage is especially clear on Region and Instance goals, where language constraints involve reference landmarks, context, and local relations.'

LANDER-Nav improves across all goal levels, with especially clear gains on Region and Instance tasks that depend more on context, reference landmarks, and local relations.

Method	Object SR	Object SPL	Room SR	Room SPL	Region SR	Region SPL	Instance SR	Instance SPL	Overall SR	Overall SPL	SeqSR	SeqSR-4
PSL (Sun et al. 2024)	–	–	–	–	–	–	–	–	–	–	–	–
SenseAct-Mono (Khanna et al. 2024)	–	–	–	–	–	–	–	–	–	–	–	–
3D-Mem-3B (Yang et al. 2025)	–	–	–	–	–	–	–	–	–	–	–	–
3D-Mem-7B (Yang et al. 2025)	–	–	–	–	–	–	–	–	–	–	–	–
Uni-NaVid (Zhang et al. 2025)	–	–	–	–	–	–	–	–	–	–	–	–
MTU3D (Zhu et al. 2025)	52.6	34.5	48.3	27.8	38.1	23.4	27.8	14.3	42.0	25.3	2.3	11.2
LANDER-Nav	55.5	37.5	49.6	31.1	43.2	26.8	33.3	18.9	45.5	28.6	3.9	13.1

Table 1: Main results on the LangMap detailed setting. Object, Room, Region, Instance, and Overall SR/SPL are reported in percentage form, together with SeqSR and SeqSR-4 in the multi-goal setting.

This result is consistent with the method design: the landmark schema preserves target, anchor, context, and relation, allowing exploration, entity binding, and stopping-viewpoint selection to operate around the same language structure. This gives the agent a deeper semantic understanding of multi-granularity tasks and improves navigation success.

The multi-goal metrics show the same trend. LANDER-Nav obtains the best SeqSR and SeqSR-4 among evaluated methods, indicating that three-stage grounding benefits sequential target completion rather than only a single local target decision.

Component Ablation Analysis

LangMap Ablation Table 2 progressively adds the three grounding components under the same LangMap multi-goal evaluation setting. Evidence Grounding corresponds to schema-guided frontier orientation, Entity Grounding corresponds to landmark-anchored object binding, and Endpoint Grounding corresponds to Landmark-Ray Endpoint Search.

Table 2 shows that the three components address different capability boundaries. Endpoint Grounding more directly improves SPL, indicating that the executable stopping viewpoint affects whether the agent can end at a suitable position after seeing and reaching the target. Entity Grounding is more critical for Region and Instance goals, emphasizing semantic understanding and decisions at more specific goal levels. Evidence Grounding gives a more moderate standalone gain, but it can push exploration toward regions that are more likely to complete schema evidence before the target appears. The Full setting obtains the highest Overall and Instance results, showing that the three components are complementary: first complete evidence, then bind the entity, and finally land at a verifiable endpoint.

OVON Ablation Table 3 evaluates the same components on HM3D-OVON. Unlike LangMap, OVON mainly provides category-level goals, so anchor and relation fields are often empty or weak.

On OVON, the full model also achieves the best Overall SR/SPL among the evaluated variants, indicating that landmark-schema grounding remains compatible with category-level open-vocabulary search. The gains mainly come from Evidence and Endpoint: the former helps unseen-category targets enter the observable region earlier, and the latter improves reachability and stopping stability. The standalone gain of Entity Grounding is smaller, which matches the instruction format of OVON; when the goal only pro-

vides a category name, there is no available evidence for anchor/relation binding.

Schema Evidence Richness Analysis

Table 5 compares detailed and concise descriptions on Region and Instance goals in LangMap. Concise descriptions usually keep only the target phrase, while detailed descriptions additionally include visual attributes, nearby objects, and scene context.

VLM Backbone Sensitivity

To analyze the influence of the VLM backbone, we conduct a controlled evaluation on a 5% shard of the multi-goal evaluation tasks.

Table 4 shows that the VLM backbone has a clear influence on the overall navigation performance. `qwen-vl-plus` achieves the best SR and SPL on this shard, outperforming `gpt-4o-mini` by 2.00 percentage points in SR and 1.71 percentage points in SPL. This indicates that stronger visual-language grounding can improve both target-reaching success and path efficiency within the same navigation framework.

`gpt-4o-mini`, which is used as the default VLM in the main experiments, ranks second overall and remains close to the best-performing model. The small gap between `gpt-4o-mini` and `qwen-vl-plus` suggests that the proposed pipeline does not depend on a single VLM, while still benefiting from stronger backbone-level semantic and visual reasoning. `qwen3-vl-8b-instruct` obtains a comparable SR to `gpt-4o-mini`, but its lower SPL suggests that it may lead to less efficient trajectories or less stable intermediate decisions under the same grounding pipeline. `gemini-2.5-flash` obtains the lowest SR and SPL in this run.

Overall, these results suggest that LANDER-Nav is compatible with multiple VLM backbones, but the quality of the VLM still affects the reliability of schema parsing and semantic evidence scoring. Since all variants share the same navigation pipeline and only replace the VLM backbone, the comparison indicates that future improvements in VLM grounding ability can further strengthen LANDER-Nav without changing the overall navigation architecture.

Detailed descriptions bring stable gains, especially for instance goals. They provide richer instance-level descriptions and stronger instance distinguishability, and LANDER-Nav

Components			Overall (SR / SPL)	Object (SR / SPL)	Room (SR / SPL)	Region (SR / SPL)	Instance (SR / SPL)
Endpoint	Evidence	Entity					
			42.0 / 25.3	52.6 / 34.5	48.3 / 27.8	38.1 / 23.4	27.8 / 14.3
✓			43.6 / 27.9	55.4 / 36.0	49.6 / 30.7	41.5 / 29.1	27.2 / 14.7
	✓		42.7 / 26.3	53.3 / 35.6	49.4 / 29.3	39.1 / 24.8	28.6 / 15.2
		✓	44.4 / 27.3	52.0 / 32.7	47.7 / 29.2	44.0 / 28.5	33.2 / 18.0
	✓	✓	44.3 / 27.4	54.1 / 35.9	50.4 / 30.1	42.3 / 27.2	29.6 / 15.8
✓	✓	✓	45.5 / 28.6	55.5 / 37.5	49.6 / 31.1	43.2 / 26.8	33.3 / 18.9

Table 2: Component ablation on LangMap. SR/SPL is reported in percentage form. Each row indicates which grounding components are enabled.

Components			Val Seen SR	Val Seen SPL	Val Seen Syn. SR	Val Seen Syn. SPL	Val Unseen SR	Val Unseen SPL	Overall SR	Overall SPL
Endpoint	Evidence	Entity								
			53.3	18.6	43.3	12.5	39.6	12.2	45.4	14.4
✓			55.8	20.2	46.7	13.2	41.3	13.7	47.9	15.7
	✓		54.8	19.7	45.3	13.8	42.0	13.2	47.4	15.6
		✓	53.8	18.9	44.1	12.8	40.1	12.0	46.0	14.6
	✓	✓	55.0	20.9	45.6	14.2	42.1	13.4	47.6	16.2
✓	✓	✓	57.5	21.1	46.7	14.5	42.2	14.1	48.8	16.6

Table 3: OVON component ablation. Each row indicates which grounding components are enabled. SR/SPL on Val Seen, Val Seen Synonyms, Val Unseen, and Overall.

VLM Backbone	SR	SPL
qwen-vl-plus	62.0	43.7
gpt-4o-mini	60.0	42.0
qwen3-vl-8b-instruct	59.0	40.1
gemini-2.5-flash	56.0	39.9

Table 4: VLM backbone sensitivity on the multi-goal evaluation tasks. All variants use the same LANDER-Nav pipeline and differ only in the VLM backbone. SR/SPL are reported in percentage form.

Method	Description Form	Region SR	Region SPL	Instance SR	Instance SPL
LANDER-Nav	Concise	41.92	26.02	31.05	17.35
LANDER-Nav	Detailed	43.17	26.84	33.29	18.87

Table 5: Comparison between detailed and concise descriptions. SR/SPL is reported in percentage form.

effectively uses these rich cues to improve navigation performance.

Conclusion

This work introduced LANDER-Nav, a landmark-schema grounding framework for multi-granularity open-vocabulary navigation. LANDER-Nav integrates schema parsing, evidence-guided exploration, landmark-anchored entity binding, and endpoint selection in a unified embodied decision process. This enables the agent to preserve target, anchor, context, and relation roles across time and viewpoints, instead of relying on a transient sentence-level query. Exten-

sive experiments on LangMap and HM3D-OVON demonstrate improved multi-granularity goal grounding, stronger performance on Region and Instance goals, and consistent compatibility with category-level open-vocabulary navigation. These results support the idea that fine-grained embodied navigation requires persistent role-level grounding as opposed to single-query target retrieval.

Limitations and future work. LANDER-Nav relies on external perception and language grounding modules, where inaccurate object masks, missed landmarks, or incorrect schema parsing can lead to errors in memory association, target binding, and stopping-point selection. The current validation is also conducted mainly in photorealistic and static simulators with relatively reliable RGB-D observations, pose estimates, and navigability assumptions. Future work will integrate stronger perception and relation verification into the navigation loop, address sensor noise and real-time constraints during real-world deployment, and explore dynamic scenes where objects, landmarks, and occlusions may change during long-horizon navigation.

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